Anas Mohammad Ramadan Ahmed AlSobeh

PERSONAL INFORMATION



Anas AlSobeh, Ph.D.
Assistant Professor, IEEE Senior Member, ACM Member |ACM SIGHPC|ACM SIGMOD| ACM SIGAI, WiCyS Member
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SUMMARY

Dr. Anas M. R. AlSobeh is an Assistant Professor of Applied AI in the department of Information Systems & Technology (IS&T) at Utah Valley University (UVU) and formerly served as an Assistant Professor and Program Coordinator of Information Technology & Cybersecurity at Southern Illinois University Carbondale (R1). He brings over 10 years of academic and research experience across the United States, Jordan, and Bahrain. He earned a Ph.D. in Computer Science from Utah State University and holds M.Sc. and B.Sc. degrees in Computer Information Systems from Yarmouk University. Dr. AlSobeh's research spans AI-driven cybersecurity, trustworthy and explainable AI, software engineering, cloud computing, and data science, with applications in malware and adversarial robustness, media forensics, and public-interest systems. He is the Principal Investigator on a USDA-funded project (2024–2026) focused on rapid Salmonella detection in onions using microscopic imaging and AI. He has published 50+ peer-reviewed papers in leading venues (Q1/Q2) and received the IACIS 2024 Best Paper Award for work on deepfake detection. Beyond academia, he has contributed to and led international initiatives funded by EU Horizon 2020, Erasmus+, GIZ, DAAD, and Nuffic in areas including refugee education, health/public-interest data systems, and cybersecurity capacity building. He has also served as an expert consultant with ICMPD, supporting training and applied analytics in migration contexts (AI and statistical analysis for migration). Dr. AlSobeh is an IEEE Senior Member, ACM SIGHPC member, and active mentor/reviewer with service roles across major journals and conferences (e.g., IEEE Access, Springer Nature venues, MDPI Systems, and other peer-reviewed outlets). His professional training/certifications include NVIDIA "Building Transformer-Based NLP Applications," AWS Cloud Practitioner, and MIT's Data Science and Machine Learning program.

RESEARCH INTERESTS

Software Engineering, Cybersecurity Analysis, Artificial Intelligence and Machine Learning (AI/ML), High-performance computing (HPC), Data Analysis, Web Technology, Health Information Systems, Aspect-Oriented Programming (AOP), Distributed Systems, and Computing Vision.

TEACHING INTERESTS

Web Security, Web Design, Client/Server Programming, Java/C#/Python Programming, Software Engineering, Database Management Systems, Occupational Internship, Senior Project, Discrete Math, Visual Programming, Object-Oriented Programming, Software Design and Modeling, Data Science, Machine Learning, AI/ML, and Human Computer interaction (HCI).

EDUCATION

January 2011– December 2015

Doctorate Degree (PhD), Utah State University, Logan, Utah, USA

Computer Science, GPA: 3.94 (95.5 Excellent)

Dissertation Title: Improving Reuse of Distributed Transaction Software with Transaction-Aware Aspects

Mentor: Stephen Clyde (swc@mdsc.com)

August 2007–January 2010

Master Degree [M.Sc], Yarmouk University, Irbid, Jordan

Computer Information Systems. GPA: 88.2 (Excellent)

September 2003– January 2007

Bachelor Degree [B.Sc], Yarmouk University, Irbid, Jordan

Computer Information Systems, GPA: 86.2 (Excellent)

WORK EXPERIENCE

1 JANUARY 2026 – PRESENT

Assistant Professor, Applied AI, Département of Information Systems & Technology (IS&T), Utah Valley University, Orem, Utah, USA.

As an Assistant Professor of Information Systems/Applied AI at UVU, my job duties involved a balance of teaching, research and service responsibilities. I conduct research in my area of expertise, deliver courses to undergraduates, and engage with the university and broader community through service activities.

Teaching (75%):

- Teaching AI, Cybersecurity, and Information Systems Courses.
- Engaging with students through active learning methods and providing academic advising to students.
- Developing new course materials (In-person, hybrid, and online) and revising existing ones to keep up with the latest advances in the field.

Research (15%):

- Conducting research in the field of Applied AI and related areas, and developing and submitting research proposals to funding agencies.

Service (10%):

- Serving on departmental, college, and university committees (e.g., assessment, accreditation, searching)
- Participating in outreach activities to engage with the community and promote the department's mission.
- Provide AI workforce training.

1 JANUARY 2023 – 31 DECEMBER 2025

Assistant Professor, Information Technology, School of Computing, Southern Illinois University, Carbondale, Illinois, USA.

ITEC Program Coordinator

As an Assistant Professor of Information Technology at Southern Illinois University, my job duties involved a balance of research, teaching, and service responsibilities. I conduct research in my area of expertise, deliver courses to undergraduates, and engage with the university and broader community through service activities.

Research (30%):

- Conducting research in the field of IT and related areas, and developing and submitting research proposals to funding agencies.

Teaching (50%):

- Engaging with students through active learning methods and providing academic advising to students.
- Developing new course materials (In-person and online) and revising existing ones to keep up with the latest advances in the field.

Service (20%):

- Coordinating the Information Technology (ITEC) and Cybersecurity (CTEC) Program Programs.
- Preparing course schedule per Academic Year (AY)
- Serving on departmental, college, and university committees (e.g., assessment, accreditation, searching)
- Participating in outreach activities to engage with the community and promote the department's mission.

11 JULY– 31 AUGEST 2025 **Expert consultant in course development**

Designed two EQF/MQF Level-4 vocational courses for ICMPD's MCP Med TI in 2025—Migration and Artificial Intelligence and Statistical Analysis in a Migration Context—producing the full package of syllabus and agenda, standardized lesson plans with practical exercises and handouts, a self-study methodology (core and supplementary reading lists), and assessment mechanisms including a final exam.

SEPT 2022- DECEMBER 2022 **Associate Professor, College of Information Technology, Kingdom University, Bahrain.**

As an academic professional, I was responsible for teaching an IT course and developing a curriculum study plan and course specifications for an IS master program that includes two tracks, namely cybersecurity and data analysis. I was also committed to engaging students in various IT-related activities, including organizing cybersecurity workshops in collaboration with the Department of Security in Bahrain. Additionally, as a representative of the university on formal occasions.

28 APRIL 2020 – SEPT 2022 **Associate Professor, Information Systems Department, Yarmouk University, Jordan.**

As an academic professional, I was responsible for various academic and administrative duties at the university. This includes delivering a requisite number of classes, providing guidance and supervision to graduate students, and actively participating in departmental meetings. I also provide academic support to professors and faculty members and assist with various departmental duties. Additionally, I am responsible for recruiting, training, and mentoring new teaching assistants and other junior staff, as well as conducting research and publishing papers in academic journals. In my role, I represent the university at conferences, delivering presentations as needed, and writing proposals to secure funding for research. I actively participate in faculty and departmental meetings, providing suggestions for improvement, and serving on various committees to continuously improve undergraduate and graduate-level curricula and training programs. With my extensive experience, I have supervised several M.Sc. projects and theses in various universities in Jordan, served on several university-level committees, and represented the department at faculty-level meetings. Additionally, I have experience as a technical team member of several

international journals, further demonstrating my expertise in the field. Overall, I am committed to making significant contributions to the success of the university, and to continuously improving both academic and administrative processes to benefit the students and faculty.

31 JANUARY 2016 – 28 APRIL 2020

Assistant Professor, Information Systems Department, Yarmouk University, Jordan.
My job duties involved conducting research, delivering courses, and engaging with the university and broader community. In addition, contributing to the success of the department and university, and helping prepare the next generation of information technology professionals. This includes developing and submitting research proposals, preparing, and delivering lectures, grading assignments, providing academic advising to students, and participating in outreach activities.

30 SEPT 2019 – 12 OCT 2021

Director of the Refugees, Displaced Persons, and Forced Migration Studies Centre, **Yarmouk University, Irbid Jordan**.
As an experienced academic leader, I'm responsible for developing and fostering collaborations between academia and organizations to support refugees, displaced persons, and forced migrants. This involves using innovative research methods to provide cutting-edge solutions to meet the needs of these populations. Additionally, I leverage the expertise and resources of leading research institutions to deliver research and training programs for local and international organizations. My ultimate goal is to establish a comprehensive network of partnerships to support the needs of refugees, displaced persons, and forced migrants in Jordan, with a particular focus on higher education and career opportunities.

2021-2022

Data Science Consultant for ICMPD, Jordan
Leading a local team of researchers from Jordan to work on the latest assessment needs of migrants in Jordan and developing custom solutions to support the implementation of the Mobility Partnership (MP) between the EU and the Hashemite Kingdom of Jordan. In addition, I'm responsible for leading and training government staff on the latest data analysis technologies to provide cutting-edge solutions to real-world challenges. So, I designed custom training programs for policymakers and technical staff in the areas of data analysis. The project is "Strengthening the Evidence Base for Migration Policies: A Central Migration Data Management Solution for Jordan (MIDAM)), funded by the Danish Ministry of Foreign, Affairs and implemented by the International Centre for Migration Policy Development (ICMPD).

JAN 2011- DEC 2015

Research Assistant & Teaching Assistant
I worked closely with Dr. Clyde on research projects and assisted with ongoing teaching activities in the computer science department.
As a research assistant with Dr. Stephen Clyde, my duties included:

1. Conducting research in the field of distributed transaction systems and assisting with ongoing research projects ECCS.
2. Attending research meetings and conferences with Dr. Clyde.
3. Providing general support to Dr. Clyde in his work and research activities.

As a teaching assistant with the computer science department, your duties may include:

1. Providing support to students in the classroom and during office hours.
2. Grading assignments, quizzes, and exams.
3. Providing feedback to students on their work and assisting with academic advising.

4. Assisting with lab activities and providing technical support to students.

JAN 2010– DEC 2011 Full Time Lecturer, Computer Information Systems, Yarmouk University, Irbid, Jordan.

May 2007– Jan 2010 Full-Time Teacher, Ministry of Education, Elementary and Secondary School Bayt-Yafa for Children, Irbid, Jordan.

May 2006- 2007 Software .Net Developer, Aman Information Systems Company, Amman, Jordan.

PUBLISHED JOURNALS AND CONFERENCE PAPER

Selected Referred Published Journal Papers

- **Anas AlSobeh**, Abhi D. Rajagopala, Izzat Alsmadi, Bilal Al-Ahmad, Amer AbuGhazaleh, Namariq Dhahir, Malek Rababa, “BactoDetect: Real-Time Bacterial Colony Detection on Heterogeneous HPC Architectures for Scalable Food Safety Monitoring”, **Submitted** to Supercomputing, Scopus Q1, 2026.
- Almousa, Omar, Yahya Tashtoush, **Anas AlSobeh**, Plamen Zahariev, and Omar Darwish. 2026. "SiAraSent: From Features to Deep Transformers for Large-Scale Arabic Sentiment Analysis" *Big Data and Cognitive Computing* 10, no. 2: 49. https://doi.org/10.3390/bdcc10020049I, Scopus Q1. 2026.
- Md Faisal Kabir, Abhishek Mandar Mahakal, Yucheng Wang, **Anas AlSobeh**, Bilal Al-Ahmad, Rami Alkhawaldah (2025). “Proximity-Constrained Counterfactuals for Explainable Diabetic Risk Assessment", submitted to "Applied Computational Intelligence and Soft Computing", Wiley, Scopus Q1.
- **AlSobeh, Anas**, Amani Shatnawi, and Aws Magableh. 2025. "AspectFL: Aspect-Oriented Programming for Trustworthy and Compliant Federated Learning Systems" *Information* 16, no. 12: 1048. https://doi.org/10.3390/info16121048I, Scopus Q2.
- Omar Darwish, Shorouq Al-Eidi, Abdallah Al-Shorman, Majdi Maabreh, **Anas Alsobeh**, Yahya Tashtoush, Plamen Zahariev, LinguTimeX: A Framework for Multilingual CTC Detection Using Explainable AI and Natural Language Processing. *CMC-Computers, Materials & Continua*, Q1, 2025.
- Hassouna, Ahmed Alaa Abdelbaky, Mohamed Badr Ismail, Amany Shaban Hassan, Huthaifa Ashqar, **Anas MR Alsobeh**, Abdallah A. Hassan, and Mohammed Elhenawy. "Evaluating Deep Learning Change Detection in Aerial Imagery: A New Multi-Metric Benchmarking Framework." (2025). Submitted to: *International Journal of Aerospace Engineering*, *Wiley*, Scopus Q1. (Preprinted).
- Al-Ahmad, Bilal, **Anas Alsobeh**, Omar Meqdadi, and Nazimuddin Shaikh. "A Student-Centric Evaluation Survey to Explore the Impact of LLMs on UML Modeling." *Information* 16, no. 7 (2025): 565. Scopus Q2.
- Terawi, Nima, Huthaifa I. Ashqar, Omar Darwish, **Anas Alsobeh**, Plamen Zahariev, and Yahya Tashtoush. "Enhanced Detection of Intrusion Detection System in Cloud Networks Using Time-Aware and Deep Learning Techniques." Computers 14, no. 7 (2025): 282. Scopus Q1.
- Al-Shawakfa, E. M., **Alsobeh, A. M. R.**, Omari, S., & Shatnawi, A. (2025). RADAR#: An Ensemble Approach for Radicalization Detection in Arabic Social Media Using Hybrid Deep Learning and Transformer Models. *Information*, *16*(7), 522. https://doi.org/10.3390/info16070522. Scopus Q2.

- Ahmed Al-Sharifin, Noor Bahr, Amani Shatnawi, **Anas M.R AlSobeh** (2025). The Effectiveness of a Psychosocial Support-Based Program in Raising Legal Empowerment among Refugees in Jordan, فاعلية برنامج يستند إلى الدعم النفسي الاجتماعي في رفع مستوى التمكين القانوني لدى اللاجئين في الأردن، . Dirasat: Human and Social Sciences, University of Jordan, Vol. 52, No. 1 (2025): 18-31 .**(In Arabic, Q3, Published, 2025)**
- **Anas Alsobeh**, Amani Shatnawi, Bilal Al-Ahmad, Alhan Aljmal, Samer Khamaiseh, (2024). AI-Powered AOP: Enhancing Runtime Monitoring with Large Language Models and Statistical Learning. International Journal of Advanced Computer Science and Applications (IJACSA). **(Scopus Q3 -SJR H-Index: 74, published, 2024)**.
- Alazzam, I., **AlSobeh, A. M. R.,** & Bani Melhem, B. (2024). Enhancing integration testing efficiency through AI-driven combined structural and textual class coupling metric. *Online Journal of Communication and Media Technologies*, 2024, 14(4), e202460 e-ISSN: 1986-3497. **(ISI, Q1, published, 2024)**.
- **AlSobeh, Anas MR**, Khalid Gaber, Mahmoud M. Hammad, Maryam Nuser, and Amani Shatnawi. "Android malware detection using time-aware machine learning approach." *Cluster Computing* (2024): 1-22. **(ISI, Q1, published, 2024)**.
- Alshattnawi, S.; Shatnawi, A.; **AlSobeh, A.M.R**.; Magableh, A.A. Beyond Word-Based Model Embeddings: Contextualized Representations for Enhanced Social Media Spam Detection. Appl. Sci. 2024, 14, 2254. https://doi.org/10.3390/app14062254. **(ISI, Q1, published, 2024)**.
- M. Alfreihat, O. Almousa, Y. Tashtoush, **A. AlSobeh**, K. Mansour and H. Migdady, "Emo-SL Framework: Emoji Sentiment Lexicon Using Text-Based Features and Machine Learning for Sentiment Analysis," in IEEE Access, doi: 10.1109/ACCESS.2024.3382836.
- Ahmad Radwan, Mohannad Amarneh, Hussam Alawneh, Huthaifa I. Ashqar, **Anas AlSobeh**, Aws Magableh. (2024). Predictive Analytics in Mental Health: Leveraging LLM Embeddings and Machine Learning Models for Social Media Analysis. International Journal of Web Services Research (IJWSR). **(ISI, Q1, published, 2024)**.
- Ahmed Al-Sharifin, Mu'ayyad Megdadi, Amani Shatnawi, **Anas AlSobeh**, Aya Akkawi. (2024). The Effectiveness of a Training Program based on health education in improving Health Empowerment level among Refugees in Jordan. Dirasat: Human and Social Sciences. (**Q3**, **published , 2023**)
- **M. R. AlSobeh** and A. A. Magableh, (2023) "BlockASP: A Framework for AOP-Based Model Checking Blockchain System," in IEEE Access, vol. 11, pp. 115062-115075, 2023, doi: 10.1109/ACCESS.2023.3325060.
- **AlSobeh, A. M. R.** (2023). OSM: Leveraging model checking for observing dynamic behaviors in aspect-oriented applications. Online Journal of Communication and Media Technologies, 13(4), e202355. https://doi.org/10.30935/ojcmt/13771
- **AlSobeh, A. M. R.**, AlAzzam, I., Shatnawi, A. M. J., & Khasawneh, I. (2023). Cybersecurity awareness factors among adolescents in Jordan: Mediation effect of cyber scale and personal factors. Online Journal of Communication and Media Technologies, 13(2), e202312. https://doi.org/10.30935/ojcmt/12942
- **Alsobeh, A.** ., & Aloudat , A. . (2022). The Repercussions of the COVID-19 Pandemic on Higher Education and its implications for Syrian Refugees Students (An Analytical Descriptive Study). Dirasat: Human and Social Sciences, 49(5), 150–166.

https://doi.org/10.35516/hum.v49i5.2770s

- Shatnawi, A. M., **AlSobeh, A**. M., AL-Mifleh, E. I., & Migdady, A. F. (2022). The Effectiveness of a Program Based on Psychosocial Support in Raising the Level of Family Empowerment among Refugees in Jordan. International Journal of Psychological and Educational Research, 1(4).
- AL-Refai, Nader S., Noor A. Albahr, Amna F. Migdady, Eman I. AL-Mifleh, Amani Shatnawi, and **Anas AlSobeh**. The Effectiveness of a Training Program Based on psychosocial support to Improve psychological empowerment Level among Refugees in Jordan. "فاعلية برنامج تدريبي مستند إلى الدعم النفسي الاجتماعي في رفع مستوى التمكين النفسي لدى اللاجئين." International Journal of Psychological and Educational Research 1, no. 2 (2022).
- Sawsan Alshattnawi, **Anas M.R. AlSobeh**. [2021]. A Cloud-Based IoT Smart Water Distribution Framework Utilizing BIP Component: Jordan As A Model. International Journal of Cloud Computing. Inderscience Enterprises Ltd. Scopus.
- **Alsobeh, A.** and Aloudat, A., 2022. The Repercussions of the COVID-19 Pandemic on Higher Education and its implications for Syrian Refugees Students (An Analytical Descriptive Study). Dirasat: Human and Social Sciences, 49(5), pp.150-166. DOI: https://doi.org/10.35516/hum.v49i5.2770. Scopus.
- Amin Jarrah, Abedalmuhdi Almomany, **Anas M.R. AlSobeh** and Eman Alqudah, [2021]. High-Performance Implementation of Wideband Coherent Signal-Subspace (CSS) based DOA Algorithm on FPGA. *Journal of Circuits, Systems, and Computers*. World Scientific Publishing Company. ISI.
- **AlSobeh, A. M.,** AlShattnawi, S., Jarrah, A., & Hammad, M. M. (2020). Weavesim: A Scalable and Reusable Cloud Simulation Framework Leveraging Aspect-Oriented Programming. *Jordanian Journal of Computers and Information Technology (JJCIT)*, 6(02). (ISI)
- Aws A. Magableh, **Anas M. R. Alsobeh**, Ahmad F. Klaib, (2019). An Evaluation of the Usage of Aspect Orientation and the Gap Between Academic Research and Industry Needs. *Journal of Theoretical and Applied Information Technology*. ID: 39546-JATIT Scopus. (Little Lion Scientific, Scopus).
- Sawsan Alshattnawi, Iyad Abu Doush, **Anas Alsobeh**, (2019). Online Mobile-Based Personal Tour Application Using Artificial Bee Colony Optimization Algorithm. *Journal of Theoretical and Applied Information Technology*. (Scopus).
- Malek M. Barhoush, Najib A. Kofahi, Khalid M.O. Nahar, **Anas M.R. Alsobeh**, Ameera Jaradat, Bayan Alomari, (2019). Performance Enhancement of the Advanced Encryption Standard Via Pipelined Implementation. *Journal of Theoretical and Applied Information Technology*. 15th August 2019. Vol.97. No 15, pp. 4213-4226. (Scopus)
- **Anas M.R. AlSobeh**; Rafat Hammad; Abdel-Karim Al-Tamimi, (2019). A Modular Cloud-based Ontology Framework for Context-aware EHR Services. *International Journal of Computer Applications in Technology*, 2019 Vol.60 No.4, pp.339 – 350. DOI: 10.1504/IJCAT.2019.101181. (Inderscience, Scopus).
- **Anas M.R. AlSobeh**; Ahmad F. Klaib; Ahlam AlYahya, (2019). A National Framework for E-health Data Collection in Jordan with Current Practices. *International Journal of Computer Applications in Technology (IJCAT)*, 2019 Vol.59 No.1, pp.64 – 73. https://doi.org/10.1504/IJCAT.2019.097118 (Inderscience, Scopus).

- **Anas M.R. Alsobeh**, Aws Abed Al Raheem Magableh, Emad M. AlSukhni, (2018). Runtime Reusable Weaving Model for Cloud Services Using Aspect-Oriented Programming: The Security-Related Aspect. *International Journal of Web Services Research*, Volume 15 Issue 1, January 2018, Pages 71-88, IGI Publishing Hershey, PA, USA. DOI: 10.4018/IJWSR.2018010104. (IGI-Global, Thomson Reuters-ISI, Impact: 0.667).
- **Anas M. R. AlSobeh**, and Aws A. Magableh, (2018). Architectural Aspect-Aware Design for IoT Applications: Conceptual Proposal. *International Journal of Computer Science & Information Technology (IJCSIT)* Vol 10, No 6, pp. 1-8, December 2018.
- Aws A. Magableh, and **Anas M. R. AlSobeh,** (2018). Securing Software Development Stages Using Aspect-Orientation Concepts. *International Journal of Software Engineering & Applications (IJSEA)*, Vol.9, No.6, pp. 57-71, November 2018. (Scientific Research Publishing, ISI)
- **Anas M. R. AlSobeh**, Stephen W. Clyde, (2016). TransJ: An Abstract Independent-Framework for Weaving Crosscutting Concern into Distributed Transactions. *Journal of Computer Technology and Application (计算机技术与应用: 英文版,*. Issue 4 Vol.7 April 2016, pages 173-195, doi: 10.17265/1934-7332/2016.04.001.
- Al-Kabi, M., Wahbeh, A., **Alsobeh, A.**, Aleroud, A., & Alsmadi, I. (2012). Examining web search trends across Arab countries. *Arabian Journal for Science and Engineering*, *37*(6), 1585-1593. (Springer, Thomson Reuters (ISI), Scopus)
- **AlSobh, A.,** Al Oroud, A., Al-Kabi, M. N., & AlSmadi, I. (2010). A multilingual and location evaluation of search engines for websites and searched for keywords. *Brazilian Journal of Information Science: research trends*, 4(1).

Selected Conference Papers

- **Anas AlSobeh**, "Beyond Trust: An Evidence-Based Benchmark for AI Decision Readiness in High-Stakes Operational Environments." *66th Annual IACIS Conference*. Accepted; under revision for Issues in Information Systems (IIS).
- **Anas AlSobeh**, Amani Shatnawi, "From Transcript to Trusted Portfolio: A Provenance-Centered Model for AI and Cybersecurity Transfer Pathways." *66th Annual IACIS Conference*. Accepted; under revision for Issues in Information Systems (IIS).
- **Anas AlSobeh**, "Applied AI for Fair Mortgage Lending: A Fairness-Constrained Gradient Harmonization Framework." *66th Annual IACIS Conference*. Accepted; under revision for Issues in Information Systems (IIS).
- **Anas AlSobeh**, "LLM-Enhanced Ensemble Learning for Android Malware Analysis and Zero-Day Intrusion Detection." *ICFT 2026*. Conditionally Accepted.
- **Anas AlSobeh**, Amani Shatnawi, Subhanjan Bikram, "Teaching Exposure Management Without Production Data: A Real-World KEV/EPSS Analysis for an AI Cybersecurity Sandbox." *35th CCSC Rocky Mountain Conference*, October 16–17, 2026, Regis University, Denver, CO. Submitted.
- **Anas MR Alsobeh**, Raneem M Alkurdi, Omar Darwish, "MIA-Bench: Quantifying the Phase Transition of User Re-Identification from Sparse Social Media Text." *Proceedings of the 17th International Conference on Information and Communication Systems (ICICS)*, pp. 1–10, 2026. https://dl.acm.org/doi/abs/10.1145/3812734.3813426
- Amani Shatnawi, **Anas AlSobeh**, Samer Khamaiseh, Muhammad Al-Abdullah, "Secure-by-Design Framework for Agentic AI: Mitigating Toxic Flows and Adversarial Exploits in Multi-Agent Ecosystems." *2026 Intermountain*

Engineering, Technology and Computing (IETC), pp. 1–6. IEEE. https://ieeexplore.ieee.org/abstract/document/11568651/

- **Anas AlSobeh**, Namariq Dhahir, Malek Rababa, Bibek Koirala, Amer AbuGhazaleh, "Rapid detection of Salmonella Typhimurium microcolonies using phase-contrast imaging and deep learning." Journal of Food Protection (2026), JFP 100934. In Press. https://doi.org/10.1016/j.jfp.2026.100934
- Malek Rababa, **Anas AlSobeh**, Namariq Dhahir, Amer AbuGhazaleh, "Backbone and Feature Fusion Design for YOLOv8-Based Bacterial Microcolony Detection in Microscopy Images." *Applied Sciences*16, no. 9: 4241. MDPI. ISI Q2. https://www.mdpi.com/2076-3417/16/9/4241
- Hanan Hashaykeh, **Anas AlSobeh**, Amani Shatnawi, Omar Darwish, "SeatLux: Explainable AI for Seat-Level Visual Comfort Prediction and Illuminance Targets in Fluorescent and LED Classrooms." *2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC)*, pp. 1–5. https://ieeexplore.ieee.org/abstract/document/11542294/
- Samer Khamaiseh, Steven Chiacchira, Aibak Aljadayah, **Anas Alsobeh**, Abdullah Al-Alaj, Omar Elariss, "ImageArmor: Simple, Yet Effective Image Immunization Against Diffusion-based Editing". *IEEE World Congress on Computational Intelligence (IJCNN)*, 21-26 June 2026 | MECC Maastricht, the Netherlands, 2026. CORE Rank B, Scopus. Accepted.
- **Anas AlSobeh**, Amani Shatnawi, and Rafat Hammad, "Translator-Oriented AI for Medical Record Reconciliation (TOAI-MR)", 2026 IEEE 2nd International Conference on Secure IoT, Assured and Trusted Computing (SATC), pp. 1–5. https://ieeexplore.ieee.org/abstract/document/11542265/
- Izzat Alsmadi and **Anas AlSobeh**" A Reproducible Framework for Institutional Domain Adaptation of LLMs via Supervised Fine-Tuning and Retrieval-Augmented Generation", The 2nd International Conference on Artificial Intelligence Systems (AIS 2026)., Texas A&M San Antonio, 2026. Accepted.
- Alsmadi, I., & **Alsobeh, A.** (2026). TAMUSA-Chat: A Domain-Adapted Large Language Model Conversational System for Research and Responsible Deployment. *arXiv preprint arXiv:2603.09992*.
- **Anas AlSobeh**, "From Transcript to Trusted Portfolio: A Provenance-Centered Model for Building Trusted AI and Cybersecurity Transfer Pathways.", 4th *BRICCs-Pathways Conference* 2026 in Alexandria, Virginia. March 15-17, 2026.
- Alsmadi, I., & **AlSobeh, A.** (2026). TAMUSA-Chat: A Domain-Adapted Large Language Model Conversational System for Research and Responsible Deployment. *arXiv preprint arXiv:2603.09992*.
- **Anas AlSobeh**, Samer Khamaiseh, Dylan Middendorf, Jack Dziubinski, "CodeLLM-Auth: Multimodal Authorship Attribution and Verification via Supervised Contrastive Code Embeddings." IEEE 50th Annual Computers, Software, and Applications Conference (COMPSAC 2026), Madrid, Spain, pp. 732–737. https://doi.org/10.1109/COMPSAC69091.2026.00100
- **Anas AlSobeh**, Soham Batra, David Nussbaum, "Do Small Language Models Verify What They Can Solve? Synthetic Derivations with Planted Errors." 40th Conference on Neural Information Processing Systems (NeurIPS 2026), Workshop: The 6th Workshop on Mathematical Reasoning and AI.
- **Anas AlSobeh**, Soham Batra, David Nussbaum, "Attributing Evaluation Awareness: Which Training Prompts and Input Cues Drive an Evaluation-Aware Provider." 40th Conference on Neural Information Processing Systems (NeurIPS 2026), Workshop: Attributing Model Behavior at Scale (ATTRIB).

- **Anas AlSobeh**, Nayef Alqahdani, Malek Barhoush, Ahmad Al-Omari, “PoC2Defense: A Benchmark for Exploit Prediction from Early Vulnerability Signals.” 2nd IEEE International Conference on Cybersecurity and AI-Based Systems (CYBER-AI 2026), Romanian-American University, Bucharest, Romania, September 22–25, 2026.
- **AlSobeh, Anas**, and Raneem Alkurdi. "A Multi-Objective Optimization Approach for Sustainable AI-Driven Entrepreneurship in Resilient Economies." *arXiv preprint arXiv:2603.08692* (2026).
- O. Karajeh, L. Abdullah, N. Alsaedi, **A. Alsobeh** and O. Darwish, "An Improved Tweet Classifier with a Hybrid Text and Graph-based Representation," *2025 International Conference on Cybersecurity and AI-Based Systems (Cyber-AI)*, Varna, Bulgaria, 2025, pp. 298-307, doi: 10.1109/Cyber-AI66431.2025.11233344.
- **Anas AlSobeh**, Raneem AlKurdi, 2025. “ResiliAI: A Multi-Agent Reinforcement Learning Framework for AL-Driven Economic Shock Resilience and Adaptive Policy Simulation”. 8th International Conference on Entrepreneurship for Sustainability & Impact ESI, 2025. (Accepted).
- Bibek Koirala, **Anas AlSobeh**, Namariq Dhahir, Amer AbuGhazaleh, “Background-Aware Instance Segmentation for Early Detection of E. coli and Salmonella in Time-Stamped Microscopy Images”, 24th International Conference on Machine Learning and Applications (AMLA), (Accepted) Dec 3-4, 2025.
- **Anas AlSobeh,** Amer AbuGhazaleh, Namariq Dhahir, Malek Rababa**,** “XAIPath: Temporal-Environmental Explainable AI Framework for Co-Contaminated Food Pathogen Detection in Microscopic Imaging”, ICPP Companion '25: 54th International Conference on Parallel Processing Companion, San Diego, CA, USA, 8-11 September 2025. CORE Rank A.
- Samer Kamiseh, Dierdre Jost, **Anas AlSobeh**, Honglu Jiang, “ADT++: Advanced Adversarial Distributional Training with Class-wise Robustness”, The 12th IEEE International Conference on Data Science and Advanced Analytics (DSAA), CORE Rank A, (Published 9-12 October 2025).
- S. Y. Khamaiseh, S. Chiacchira, **A. Alsobeh** and A. Aljadayah, "MuAE: A Mutation Testing Framework for Evaluating Autoencoders," 2025 IEEE 49th Annual Computers, Software, and Applications Conference (COMPSAC), Toronto, ON, Canada, 2025, pp. 1015-1020, doi: 10.1109/COMPSAC65507.2025.00131.
- **Anas Alsobeh**, Zahraddeen Gwarzo and Amani Shatnawi, (2025). “ShadowPlay: Engineering Defenses Against Role-Based Prompt Injection and Dependency Hallucination in LLM-Powered Development”. Cyber-AI2025 (The International Conference on Cybersecurity and AI-Based Systems). September 2025. (Accepted).
- **Anas AlSobeh**, Mustafa Rob, Kamel Rouibah, Amani Shatnawi. "Proactive Detection of Tax Fraud Using Explainable AI Techniques: A Hybrid Approach ", 65th IACIS Annual Conference | October 1 - October 4, 2025, Sheraton Sand Key Clearwater Beach, Florida. (published in IIS Journal)
- A. Shatnawi, **A. AlSobeh**, I. Alsmadi and B. Al-Ahmad, "Tailored Large Language Models for Spam Detection: From Model Customization to Benchmarking Effectiveness," 2025 5th Intelligent Cybersecurity Conference (ICSC), Tampa, FL, USA, 2025, pp. 264-271, doi: 10.1109/ICSC65596.2025.11140025.
- L. Patel and **A. Alsobeh**, "SlangLLM: Dynamic Detection and Contextual Filtering of Slang in NLP Applications," *2025 1st International Conference on Secure IoT, Assured and Trusted Computing (SATC)*, Dayton, OH, USA, 2025, pp. 1-6, doi: 10.1109/SATC65530.2025.11137237.

- Frisbier, Gayle L., Omar Darwish, **Anas Alsobeh**, and Abdallah Al-shorman. "Identifying the Origins of Business Data Breaches Through CTC Detection." In International Conference on Network and System Security, *Network and System Security*, Springer, pp. 387-406. 2024.
- **Anas AlSobeh**, Alan Franklin, Belle Woodward, (2024). (**Best Paper Award**) "Unmasking Media Illusion: Analytical Survey of Deepfake Video Detection and Emotional Insights." The 64th IACIS Annual Conference - October 2 - 5, 2024, at One Ocean Resort and Spa, Atlantic Beach, Florida. (Oct 2024, IIS Journal, Best Paper Award)
- F. A. Hasan, H. I. Ashqar, **A. AlSobeh** and O. Darwish, "Blockchain-Based National Digital Identity Framework – Case of Palestine," *2024 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS)*, Dubrovnik, Croatia, 2024, pp. 76-83, doi: 10.1109/ICCNS62192.2024.10776538. Scopus.
- A. Harasees, B. Al-Ahmad, **A. Alsobeh** and A. Abuhussein, "A Secure IoT Framework for Remote Health Monitoring Using Fog Computing," 2024 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS), Dubrovnik, Croatia, 2024, pp. 17-24, doi: 10.1109/ICCNS62192.2024.10776425. Scopus.
- Abanoub Kased, Rana Rabee, Akram Fahmy, Hussein Mohamed, Marco Yacoub, Mohammed Elhenawy, Huthaifa I. Ashqar, **Anas Alsobeh**, Mahmoud Masoud, Abdallah A. Hassan, "Vehicle Trajectory-Based Intersection Classification Using Deep Transfer Learning," *2024 International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS)*, Dubrovnik, Croatia, 2024, pp. 158-165, doi: 10.1109/ICCNS62192.2024.10776444. Scopus.
- Darwish, O., Al-Shorman, **A., AlSobeh**, A., Tashtoush, Y. (2024). A Survey Analysis of Internet of Things (IoT) Education Across the Top 25 Universities in the United States. In: Barolli, L. (eds) Advanced Information Networking and Applications. AINA 2024. Lecture Notes on Data Engineering and Communications Technologies, vol 203. Springer, Cham. https://doi.org/10.1007/978-3-031-57931-8_47
- **Anas AlSobeh**, Belle Woodward, AI as a Partner in Learning: A Novel Student-in-the-Loop Framework for Enhanced Student Engagement and Outcomes in Higher Education, SIGITE '23, October 11–14, 2023, Marietta, GA, USA, ACM ISBN 979-8-4007-0130-6/23/10. DOI: https://doi.org/10.1145/3585059.3611405. CORE Rank: Local USA, Scoups.
- **Anas AlSobeh**, Amani Shatnawi, Integrating Data-Driven Security, Model Checking, and Self-Adaptation for IoT Systems using BIP Components: A Conceptual Proposal Model. International Conference on Advances in Computing Research (ACR'23), May 8-10, 2023, Orlando, Florida, USA. Springer Book Series: Lecture Notes in Networks and Systems (LNNS). (Scopus, 2023)
- Barhoush, M. M., **AlSobeh, A. M.**, & Al Rawashdeh, A. (2019, April). A Survey on Parallel Join Algorithms Using MapReduce on Hadoop. *In 2019 IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology (JEEIT)*. IEEE. (IEEE conference). *(pp. 381-388)* Amman, Jordan, April 2019. DOI: 10.1109/JEEIT.2019.8717427. Scopus.
- **Anas M. R. AlSobeh**, Aws A. Magableh, (2018). An Aspect-Oriented with BIP Components for Better Crosscutting Concerns Modernization In IoT Applications. *5th International Conference on Computer Science and Engineering (CSEN-2018)* Nov 25 ~ 26, 2018, Dubai, UAE, pp 21-31 At: Dubai DOI: 10.5121/csit.2018.81203. ISBN: 978-1-

921987-90-8.

- Aws A. Magableh, **Anas M. R. AlSobeh**, (2018). Aspect-Oriented Software Security Development Life Cycle (AOSSDLC). *5th International Conference on Computer Science and Engineering (CSEN-2018)* , Nov 25 ~ 26, 2018, Dubai, UAE, pp33-47 At: Dubai DOI: 10.5121/csit.2018.81204. ISBN: 978-1-921987-90-8.
- **Anas M. R. AlSobeh**, Stephen W. Clyde, (2016). Transaction-Aware Aspects with TransJ: Initial Experiment Show Promising Improvements in Reusability of Distributed Transactions. *The Eleventh International Conference on Software Engineering Advances (ICSEA 2016)*. pp.59. Italy, Rome, July 2016. (Conference, International Academy, Research, and Industry Association (IARIA))
- **Anas M. R. AlSobeh**, Stephen W. Clyde, Unified Conceptual Model for Joinpoints in Distributed Transactions, (2014). *The Ninth International Conference on Software Engineering Advances*, *ICSEA'14*. Vol. 14, pp. 8-15. France, Nice, October 2014. ISBN: 978-1-61208-367-4. (Conference, International Academy, Research, and Industry Association (IARIA))

Submitted (Under review)

- **Anas AlSobeh**, Malek Rababah, Namriq Dhar, Amer Abu Gazaleh, 2025. From Pixels to Policy: RDM Strategies for AI-Driven Bacterial Detection in Food Safety Research. *Research Data Management (BRICCs-RDM) Conference*, the Westin Alexandria Old Town, 400 Courthouse Square, Alexandria, VA 22314. July 9-11, 2025. (On press).

On going research Projects

- Sawsan AlShattnawi, **Anas Alsobeh**, Transparency and Efficiency in Cloud Forensics: A Novel DF-CFI Approach Using Explainable AI. To be Submitted: JCIS. Scopus Q1.
- **Anas M.R. AlSobeh**, AbdelRahman Alasmar, (2024). A Hybrid Reasoning Framework to Identify Cross-Cutting Concerns Using Aspect-Oriented Techniques.
- Samer, Khamaiseh, Aibak AL-Jadayah, Steven Chiracchira, **Anas AlSobeh**. "Vanguard: Post Generation Filter for Safe Text-2-Image Diffusion Models Contents".

FUNDED RESEARCH/ PROJECTS

Principal Investigator (PI)

[2024 – 2026]

- Program: Capacity Building Grants for Non-Land Grant Colleges of Agriculture, USDA-NIFA, Grant No.2024-70001-43667 & Proposal Number: 2024-02854
  Proposal Title: "Enhancing Food Safety: Rapid Detection of Salmonella in Onions Using Microscopic Imaging and Artificial Intelligence" ($150,000.00)

2026

Principal Investigator. "AI for Every Student: Building an Applied AI Incubator for CTE Programming Education." Perkins V Grant Award, Utah Valley University. $31,902. 2026.

[2019 – 2023]

- Reshaping Attention and Inclusion Strategies for Distinctively Vulnerable People Among the forcibly displaced [RAISD]. *European Union*, *Horizon 2020 (H-2020)*. Budget: € 2,662,128.00 Euro.

[2019 - 2021]

- Psychosocial Support and Trauma work in Jordan/Certified Advanced Training on Community-Based Psychosocial Support in the Context of Forced Migration. *Deutsche Gesellschaft for Internationale Zusam-menarbeit (GIZ) GmbH*. Budget: €121,676.94 Euro.

[2018 - 2020]

- Toward Innovative Design an Interactive, Secure, Learning and Research YU Cloud Environment [YUCloud]. *Deanship of Scientific Research and Graduate Studies*, *Yarmouk University*. Budget: 43,140.00 JD, approximately $ 60,846.00 USD.

[2018 - 2019]

- Design of Smart Water Distribution System Based on Cloud Computing and Internet of Things [SWDS]. *Deanship of Scientific Research and Graduate Studies*, *Yarmouk University*. Budget: 2,000.00 JD, approximately $ 2,820.00 USD.

[2017 - 2019]

- Providing Fundamental ICT Skills for Syrian Refugees [PFISR]. *European Union*, *HOPES-MADAD*. Budget: € 53,375 Euro.

[2017 - 2019]

- Transferring E-Commerce Fundamentals for Syrian Refugees [TEFSR]. *European Union*, *HOPES-MADAD*. Budget: € 39,250.000 Euro

Co-Principal Investigator (Co-PI)

2021 - 2022

- PIANO: Pharmaceutical initiative for the medicinal plants of the Middle East in research and teaching (Pharmazeutische Initiative für die Arzneipflanzen des Nahen Ostens in Forschung und Lehre), *DAAD: Deutscher Akademischer Austauschdienst German Academic Exchange Service*, Budget: € *198,780.00* Euro.

2021 – 2022

- JUICEE – Developing Jordanian Universities' Innovation Capacities and Entrepreneurship Education, *DAAD: Deutscher Akademischer Austauschdienst German Academic Exchange Service*, Budget: € *225,150.00* Euro.

Oct. 2020 –Oct 2022

- Learning & development for a Sustainable future for refugees & host communities to promote co-existence and Socio-Economic Development (LSFR), *Orange Knowledge Programme. Tailor-Made Training / Tailor-Made Training Plus*, *Nuffic*. Budget: € 531,778.

2019 – 2022

- Neurodevelopmental CAre for Refugees [NeuCARE]. *European Union*, *Erasmus+ programme*, *Capacity Building in higher education - Joint Projects*. Budget: € 668,750.00 Euro.

THESIS/DISSERTATION/ PROJECT SUPERVISION

Theses/ Dissertations

- Malek Rababah, PhD Computer Science. Supervisor, Dissertation: "*Rapid Detection of Objects Using Artificial Intelligence*".
- Khalid Hamdan Jaber, MS of Computer Information Systems, Co- Supervisor, *Thesis Title: Context-Aware Android malware detection Model using machine learning techniques*.
- Sahar AlOmari, MS of Computer Information Systems, Co-Supervisor, Thesis Title: *Social Media Text Radicalization Detection Based on Deep Learning Techniques in Arabic language*.
- Abeer Algazo, MS Student at Yarmouk University, Main Advisor. Thesis title: *A secure EHR Framework based on Microservices and Blockchain Technologies utilizing the Fluree Data Platform*.
- Alhan Jamil, MS Student at Yarmouk University, Main Advisor. Thesis title: *Towards Practical Runtime Monitoring Approach for Aspect-Oriented Programs Using Model Checking*.
- Abdel Rahman Alasmar, MS Student at Yarmouk University, Main Advisor. Thesis title: *A Hybrid Reasoning Framework to Identifying Cross-Cutting Concerns Using Aspect-Oriented Techniques*.

POST DOCTORATE/GRADUATE COMMITTEES AND SUPERVISION

- Examiner to PhD. Thesis- GameAbove College of Engineering & Technology, Eastern Michigan University (Gayle L. Frisbie): "Identifying the Origins of Business Data Breaches through CTC Detection", 2024
- Examiner to PhD. Thesis- Department of Mathematics at SIUC (Abdulaziz A. Alshammari): "Testing with the one component partial least squares and the marginal maximum likelihood estimators", 2024.
- Invitation Letter Fulbright Program: "Dr. Muhammad Qasim to join SIUC as a Postdoctoral Researcher in the School of Computing and contributing to our ongoing research on "AI-based techniques for Predicting Student Success using Constructive Engagement Analytics". 2024.

Extracurricular experience
Supervised Student Projects

2025

2024

2020-2023

- Research Poster (2025), Student Research Mentorship Currently supervising the CyberPEACE: Advancing Equitable Privacy and Security Education through Edge Innovation and Zero Trust Technology, Students: Kromwell, Ahyoni, Denmark, Sarah and Sirgiovanni, Louie.
- Workshop "Media Literacy and Deepfake Detection: Enhancing Cognitive Skills through Adaptive Educational Interventions". **WiCyS 2025.**
- Siegelman, Joseph L, "CyberGuardian: A Gamified Approach to Cybersecurity Education with AI-Assisted Learning". Undergraduate research work. Susan C. Thrasher Research Fund.
- Iman Yousef Khasawneh, MS Student at Yarmouk University, Mian Supervisor, Project Title: *Social Engineering and Cybersecurity Awareness: Jordanian Teens as Model*.
- Ahlam Al-Yahya, MS Student at Yarmouk University, Main Supervisor. Project title: *Cancer and Diabetes Data Analysis in Jordan using a Clustering Algorithm*.
- Shorooq B. Alazzam, MS Student at Yarmouk University, Main Supervisor. Project title: *Preventing SQL Injection and XSS Attacks using Machine Learning*.
- iTeach: Mobile-based Application for Teachers, 2021.
- Etrevago: Web-based Auction Systems, 2021.
- EVS: Online election system, 2019.
- Yarmouk Online Academy, Web Application, First Semester 2019.
- Smart Traffic System, IoT-based Management System, First semester 2018
- E-Election Student Application, Web-based Application, First Semester 2018
- Ezra Bethra, Charity Web Application System, Second Semester 2017
- Mr. Dentist, Scheduling System, Mobile Application, Second Semester 2017
- Automatic Barriers System, IoT Application, Second Semester 2017

Other academic services

2026 **Potential Reviewer / Editorial Board Member**, Science of Computer Programming (Elsevier, Scopus Q1 Journal). Invited via Editorial Manager, April 2026.

2026 **New TPC/PC Memberships (May 2026)** • Technical Program Committee Reviewer, ICICoS 2026 (International Conference on Internet of Computing and Sensing), August 12–13, 2026. • Program Committee Member, Cyber-AI 2026 (International Conference on Cybersecurity and AI-Based Systems). • Reviewer, ISBCOMTECH 2026 (3rd International Conference on Intelligent Systems, Blockchain, and Communication Technologies), May 2, 2026, Mount Pleasant, USA. • Reviewer, ICMLA 2025 (24th International Conference on Machine Learning and Applications), December 2–4, 2025, Boca Raton, FL.

2025 **Chair Special Session**: "AI-Driven Threat Analysis for IoT Software Security", 1st IEEE Conference on Secure and Trustworthy Cyber-Infrastructure for IoT and Microelectronics (SaTC 2025), https://satcconf.com/index.html, and https://satcconf.com/ss.html.

Journal, Book, Conference Referees

- ICMLSC 2025 (9th International Conference on Machine Learning and Soft Computing, Tokyo, Japan). Role: Technical Program Committee Member.
- CMVIT 2025 (9th International Conference on Machine Vision and Information Technology, Maldives). Role: Technical Committee Member.
- CCEAI 2025 (9th International Conference on Control Engineering and Artificial Intelligence, Ho Chi Minh City, Vietnam). Role: Technical Committee Member.
- AAIEE 2025 (IEEE International Symposium on the Application of Artificial Intelligence in Electrical Engineering, Beijing, China). Role: Technical Committee Member.
- ICSED 2025 (7th International Conference on Software Engineering and Development, Bangkok, Thailand). Role: Technical Committee Member.
- **WiCyS (Women in Cybersecurity 2025)**: Role: Volunteer Reviewer for Research Poster Submissions, Dallas, USA.

2024
- **TV Interview, KFVS12**: SIU researchers studying how AI can prevent food poisoning https://www.kfvs12.com/2024/10/30/siu-researchers-studying-how-ai-can-prevent-food-poisoning/ and https://news.siu.edu/2024/10/102924-siu-researcher-gets-150k-to-use-ai-to-prevent-food-poisoning.php
- **NSF Panelists**.
- **STEM Trainer in AI (LLM)**, SRELO Research Program, Illinois Mathematics and Science Academy: IMSA. Summer Training program, June 17-June 28, 2024.
- (**Press**) Survey: Large numbers of students using AI without professors' knowledge, 2024. (Link) Daily Egyptian, Southern Illinois University Carbondale, March 2024.
- **Guest Editor**, Special Issue on Cybersecurity and Secure Information Systems: Challenges and Solutions in the Digital Environment- Journal Systems, MDPI. Indexed in Scopus and ISI, with an impact factor of 1.9. https://www.mdpi.com/journal/systems/special_issues/421CBLW879
- **Chair**, The 38-th International Conference on Advanced Information Networking and Applications (AINA-2024) - Workshop-DIAC-2024: The 2nd Workshop on Diversified IoT Applications of Cybersecurity (DIAC-2024). 8-16/4/2024
- **Chair**, The Third Workshop on Advanced Cybersecurity Applications in IoT and Beyond (ACAIoT2024), 24-27 Sept. 2024 | Dubrovnik, Croatia. In Conjunction With The International Conference on Intelligent Computing, Communication, Networking and Services (ICCNS2024)

2017-2019 **Academic Entrepreneurship Center of Excellence**, Yarmouk University, Member.

2017- 2019 **Chair**, Faculty of Information Technology and Computer Science, Yarmouk University, Member.

2022-2023 **Advisory Board**, Refugees, Displaced Persons, and Forced Migration Studies Center, Member.

2016- 2017 **Organizing Committee**, SIGMOD 14 Conference. Member.

2014 **Academic Reviewer**, Several Conferences and Journal (Scopus and ISI), *Reviewer*.

2017 – Present **International Association of Scientific Researcher** A scientific endowment of public benefit, *Member.*

20/8/2019 - Present

Journal, Book, Conference Referees

Selected Peer review

- International Journal of web services research: 1 review.
- Applied Sciences: 3 reviews.
- Computers: 12 reviews.
- Future Internet: 11 reviews.
- Journal of Cybersecurity and Networking: 1 review.
- Environmental Modeling & Software: 1 review.
- Sensors: 1 review.
- Journal of Cybersecurity and Privacy: 1 review.
- Mathematics: 1 review.
- Sustainability: 1 review.
- Technologies: 1 review.
- Online Journal of Communication and Media Technologies: 3 reviews.
- JCIS (Journal of Computer Information Systems) Editorial Board, Role: Editorial Board Member, Responsibility: Overseeing submissions, reviews, and journal quality.

Conference Referees

- SIGITE (Special Interest Group for Information Technology Education). Role: Author and Reviewer, USA.
- AINA (Advanced Information Networking and Applications). Role: Reviewer, Switzerland.
- IACIS (International Association for Computer Information Systems). Role: Reviewer and Presenter, USA.
- ITCC (International Conference on Information Technology and Computer Communications). Role: Technical Program Committee Member, Yokohama, Japan.

2021-2022

- **Reviewer**, "IET Computers & Digital Techniques", Scopus.
- **Member of Postgraduate Studies Committee**, Faculty of Information Technology and Computer Science.
- **Chair** of Conference for Migration and Refuge in the Arab Region: Reality-Challenges-Solutions
- **Committee Member of 4th ICOIACT 2021**, 4th International Conference on Information and Communication Technology, Indonesia, August 30, 2021, by Universitas Amikom Yogyakarta.
- **Guest Speaker**, International Conference on "Smart Sustainable Material and Technologies" during July 23-24, 2021, in Berlin, Germany online.

2020

- **Technical Committee Member** of "Blockchain Technology in Cloud Computing Challenges and Open Issues" Conference. IWCMC 2020.
- **Advisory Board** for the Project and International Relations
- **Chair of Conference Protracted Displacement Conference**: Hopes, Perspectives and Solutions? Governance – Social Dimension – Academia

2019

- **Technical Committee Member of ACAI 2019 and 2020**, International Conference on Algorithms, Computing and Artificial Intelligence, Sanya, China during December 24-26.

Editorial Boards

- **Guest Editor for a Special Issue** on Cybersecurity and Secure Information Systems, MDPI-Systems, Scopus, Q3.
- **Chair Session** Education and AI, SIGITE '23, October 11–14, 2023, Marietta, GA, USA, Oct 11-14, 2023.
- **Chair Session** of Cybersecurity and Networking and Communication, The 2023 International Conference on Advances in Computing Research (ARC'23), Orlando, USA, May 8-10, 2023.
- **Editorial Board Member**, JCIS (Journal of Computer Information Systems), Scopus. Q1,

TAUGHT COURSES

- **Bachelor Information Systems**
  - CIS 101: Computer Skills
  - IT 102: Introduction to Computing Applications
  - CIS 102: Technology and Societies
  - CIS 214: Visual Programming (C#)
  - CIS 227: Human Computer Interaction
  - ITEC 280: Discrete Math for IT
  - CIS 260: Database Management Systems
  - CIS 311: Internet Applications
  - ITEC 312: Programming II- Python
  - CIS 341: Website Design
  - CIS 382: Cloud Computing
  - ITEC 409: Java Programming
  - CIS 411: Client/Server Programming
  - JRN 442: News Website Development
  - CIS 499: Graduation Project (BSc)
  - ITEC 410: Web Security
  - ITEC 419: Occupational Internship
  - INFO 1200: Computer Programming I for IS&T
  - INFO 2200: Computer Programming I for IS&T
- **Graduate Courses**
  - CIS 642: Software Design and Modelling
  - CIS 683: Web Technology
  - CIS 699: Graduation Project (MSc)
  - IT 6740: Advanced Network Defense and Countermeasures
- **Workshops/Trainings**
  - E-Psychology Support Applications
  - Entrepreneurship
  - Digital Transformation
  - E-commerce
  - HPC
  - ML/AI

PROFESSIONAL DEVELOPMENT / TRAINING / WORKSHOPS

2026

- 2026, Program Committee Member, ICTAI 2026 (38th IEEE International Conference on Tools with Artificial Intelligence), Boca Raton, FL, November 2–4, 2026
- 2026, Program Committee Member, Rev-AI 2026 (International Conference on Revolutionary Artificial Intelligence and Future Applications)
- 2026, Program Committee Member (Tutorials and Workshops), PEARC'26 (Practice and Experience in Advanced Research Computing)
- 2026, Program Committee Member, SRC2026 (Student Research Competition)
- 2026, Track Chair (Talks Track), USRSE'26 (US Research Software Engineer Association Conference)
- 2026, Technical Program Committee Member, CITIC 2026 (Conference on Information Technology, Innovation and Computing), May 5–7, 2026 — Tracks: Artificial Intelligence & Machine Learning; Information Systems; IoT, Big Data & Blockchain; Security & Privacy; Software Engineering
- 2026, Technical Program Committee Reviewer, 2026 ICCE-Taiwan (IEEE International

Conference on Consumer Electronics – Taiwan), July 1–4, 2026

- 2026, Technical Program Committee Reviewer, ETCoS-Grace 2026 (Emerging Technologies on Computing Systems), August 6–7, 2026
- 2026, Technical Program Committee Reviewer, FET'26 (Future Engineering Technologies Conference), April 22–23, 2026
- 2026, Reviewer, IETC 2026 (IEEE International Electronics Technology Conference)
- 2026, Sub-reviewer / External Reviewer, COMPSAC 2026 (IEEE International Conference on Computers, Software and Applications)
- 2026, Reviewer, International Association for Computer Information Systems (IACIS) 2026 Conference — 66th IACIS Annual Conference; reviewed scholarly submissions for conference paper evaluation and potential publication in Issues in Information Systems (IIS)
- 2026, External Reviewer / Subreviewer, Rev-AI 2026 (International Conference on Revolutionary Artificial Intelligence and Future Applications)
- Texas A&M University, BRICC Research Data Management Conference, Alexandria, Virginia March 15–17, 2026 – Presentation.
- Women for Cybersecurity (WiCys), Academic Advisor & Mentor, Washington D.C. March 11-13, 2026. (Travel Award).
- NAIRR Classroom Pilot Expansion Conference, Computing Research Association (CRA). College of Dade in Miamai, Feb 15, 2026.
- Invited Talk Proposal Accepted, RCSI (Research Computing for Science and Innovation) 2026, Swarthmore College — "Building AI-Ready Infrastructure at a Regional Teaching University: A Phased Approach to Integrating Artificial Intelligence, High-Performance Computing, and Cybersecurity at Utah Valley University." NSF travel support awarded (up to $1,680).
- OSG School 2026 Participant, Open Science Grid — OSPool computing account approved; username: anas.alsobeh, access point: ap40.uw.osg-htc.org.
- Reviewer, 35th CCSC Rocky Mountain Conference 2026, October 16–17, 2026, Regis University, Denver, CO — assigned 2 submissions for review.
- Facilitator / Participant, "From Draft to Done – Research Support Group (Works in Progress)" Learning Circle, Utah Valley University, 2026.
- Presenter, 2nd IEEE International Conference on Cybersecurity and AI-Based Systems (CYBER-AI 2026) — presented "PoC2Defense: A Benchmark for Exploit Prediction from Early Vulnerability Signals," Romanian-American University, Bucharest, Romania, September 22–25, 2026.
- Participant, NCyTE Webinar: "Teaching AI Security: Hands-On LLM Hardening with Docker Desktop and Security Gateways," NCyTE Center, October 2, 2026.
- SoTL Institute Proposal Submitted, Utah Valley University Office of Teaching & Learning, 2026–2027 — "Structured AI Scaffolding for Novice Python Learning" (INFO 1200, Computer Programming I); themes: Making Learning Fun, Assessing Student Learning; keywords: Generative AI, Introductory Programming, Instructional Scaffolding, Debugging, Transfer of Learning; data collection Spring 2027; stipend up

to $1,000.

- Co-Editor, Research Topic: "Generative AI for Transportation Operations and Planning: Methods, Applications, and Governance," Frontiers in Future Transportation (Impact Factor 1.9, CiteScore 3.7). Submission deadline: August 31, 2026.
- UVU Faculty Innovator, Utah Valley University Faculty Institute — recognized for innovative work on AI agents in coding and debugging for programming education, 2026.

2025

- PACES Research Training Workshop (Albuquerque, 2025) — completed advanced CI/HPC training.
- ACES HPC platform — Active user (2025– present), Running AI/ML workflows for microscopy and cybersecurity projects (e.g., model training/evaluation, containerized pipelines, CI/CD to HPC).
- ACCESS Campus Champions (SIU Carbondale) — Institutional liaison (2025–) Onboarding researchers to national CI/HPC resources; training and outreach.
- Texas A&M, BRICC Research Data Management Conference, July 9–11, 2025 – Presentation
- FABRIC KNIT10 & KNIT11, Oct 14, 2025 — Travel Stipend Registration confirmed.
- **Campus Champion, ACCESS Campus Champions Program** – SIU Carbondale (Appointed 2025–), Institutional liaison for national cyberinfrastructure; support and onboarding for SIU researchers; training/outreach responsibilities.
- **Training: Building Transformer-Based Natural Language Processing (NLP) Applications**, NVIDIA
- **CI Pathways Training Program – Machine Learning & AI Track**
  NSF Award #2417789 | PSC & NCSA, Summer 2025
  Completed a competitive, cohort-based training program focused on applying cyberinfrastructure tools to Machine Learning and AI workflows. Engaged in guided sessions on cutting-edge CI frameworks, hands-on assignments, and collaborative learning, under expert mentorship.

PROFESSIONAL MEMBERSHIPS

- ACM Member, AISIG, SIGHPC, SIGMOD, ACM Membership (2026-2027)
- Campus Champion, ACCESS Campus Champions Program (2025- present)
- ACM SIGHPC Member (2025–2026)
- Women in Cybersecurity (WiCyS) member (2024-present)
- IEEE Senior Member, IEEE Membership (2023-present)

HIGHLIGHTS, HONORS, LICENSES & CERTIFICATIONS AND AWARDS

2026
- The Best Paper Award at the IEEE Conference on Secure and Trustworthy CyberInfrastructure for IoT and Microelectronics (SaTC) 2026
- Agentic Artificial Intelligence: Harnessing AI Agents to Reinvent Business, Work, and Life, LinkedIn, Issued Jan 2026
- Research Security Training Certificate, Utah Valley University, Office of Sponsored Programs, Issued May 27, 2026
- .NET MAUI for Beginners, LinkedIn Learning Certificate of Completion
- Prompt Engineering with ChatGPT, LinkedIn Learning Certificate of Completion

2025
- NSF CCMNet Travel Award — PEARC25 (July 2025), Participation included CCMNet workshop and mentoring commitments
- REACH Faculty Mentor – Research Enriched Academic Challenge Award
  Southern Illinois University, Office of the Vice Chancellor for Research
  Project: Benchmarking Integrated Detection of LLM-Generated Deepfakes Across Modalities on Social Media (Award Period: 2025–2026).
  Project received $2000 in research funding and UGA student support stipend.
- ACM-SIGHPC (Special Interest Group on High Performance Computing) Member
- **Certificate**: Building Transformer-Based Natural Language Processing (NLP) Applications
  Issued by NVIDIA, May 9, 2025
  Certification ID: btma_Dy6SHWe331LF_w71w
  Verified at: learn.nvidia.com/certificates?id=btma_Dy6SHWe331LF_w71w
- SecureAI – Certificate (2025), Loyola University Chicago.
- WiCyS Scholarship ($500) for Conference, 2025.
- KNIT10-2025, Award $1,106.00.

2024
- Best Paper Award: IACIS 2024.
- Certificate Appreciation: Research Mentor during STEM Research Academy hosted by SIU STEM Education Research Centre.

2022- 2023
- Data Science and Machine Learning: Making Data-Driven, Decisions program by MIT Institute for Data, Systems, and Society (IDSS).
- AWS Cloud Partitioner Certificate. Amazon Web Services.
- Institutional/Signatory Official: Human Subject Research, CITI program.

2017
- Oracle Academy Certificate, Training of Trainees (ToT).

2015
- Member of USU Honoured Graduate Students.

2014
- Research and Projects Grant Proposal Award, Utah State University, USA.
- Full Tuition Award, Utah State University.

2011-2015
- PhD Scholarship, Yarmouk University, Jordan.

2011- 2014
- First Rank Student Award, Yarmouk University, Jordan.

REFERENCES

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- Izzat Alsmadi, Professor, Computational, Engineering and Mathematical Sciences, Texas A&M, San Antonio, Phone: 210-784-2313, ialsmadi@tamusa.edu,
- Stephen Clyde, Associate Professor of Computer Science, Utah State University, Logan, Utah, USA. Email: swc@mdsc.com. 435-764-1569.
- Omar Darwish, PhD, Assistant Professor, Eastern Michigan University, +1(269) 254-4979, odarwish@emich.edu.
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- Samer Y. Khamaiseh, Assistant Professor, Director of the Laboratory of AI Security Research (LAiSR), Department of Computer Science & Software Engineering, Miami University, 205 Benton Hall, 510 E. High St., Oxford, OH 45056, Phone: 513-529-0340. khamaisy@miamioh.edu
- Abdel Nasser Hindawi, PhD, Program Coordinator DAAD JORDAN / Manager of the Master Program in the EDU-SYRIA III Project, DAAD Jordan. 2nd Investment Building - 4th Floor, Ahmed Al-Tarawneh Street, Jubeyha, Amman - Jordan, Mobile: 00962 795 620 679, Email: hindawi@daad.de, nasserhindawi@gmail.com. Website: www.daad-jordan.org, www.daad.de.